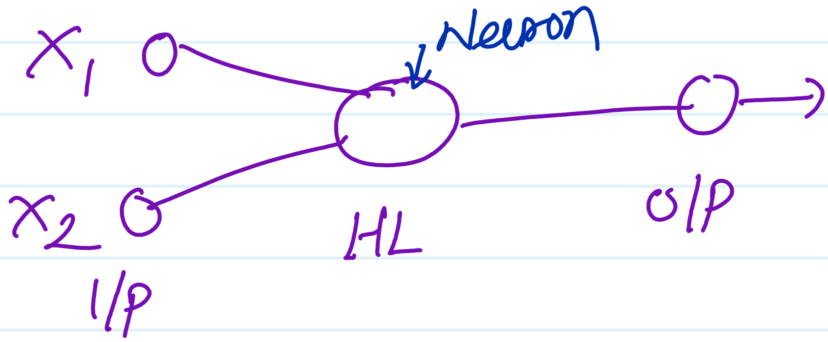
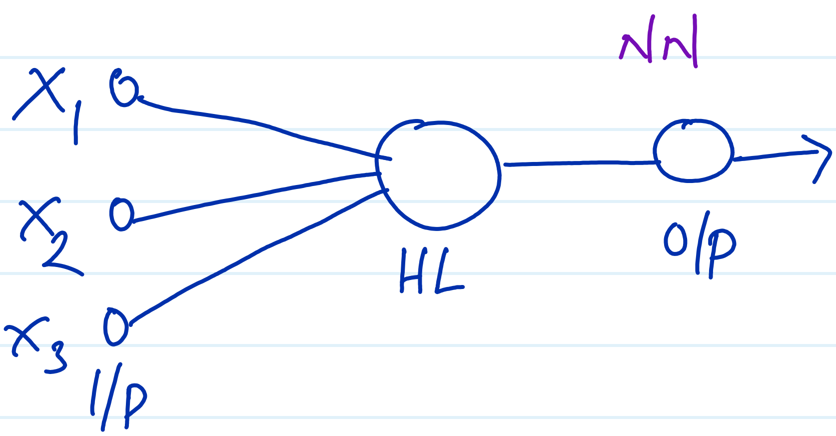
1) Neuron :- single unit

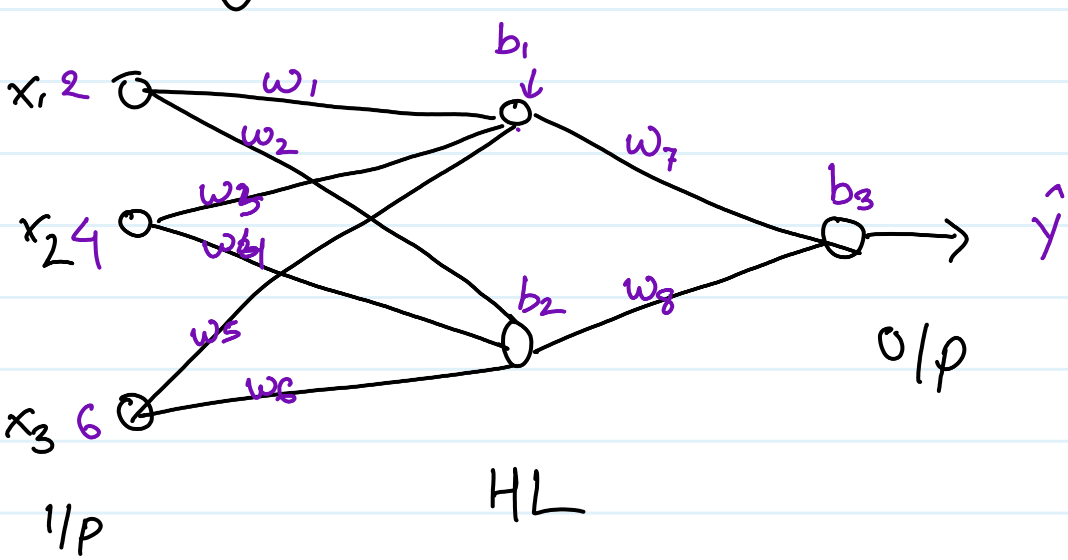
x_1 x_2 x_3



2) Perceptron :-



3) weight & bias



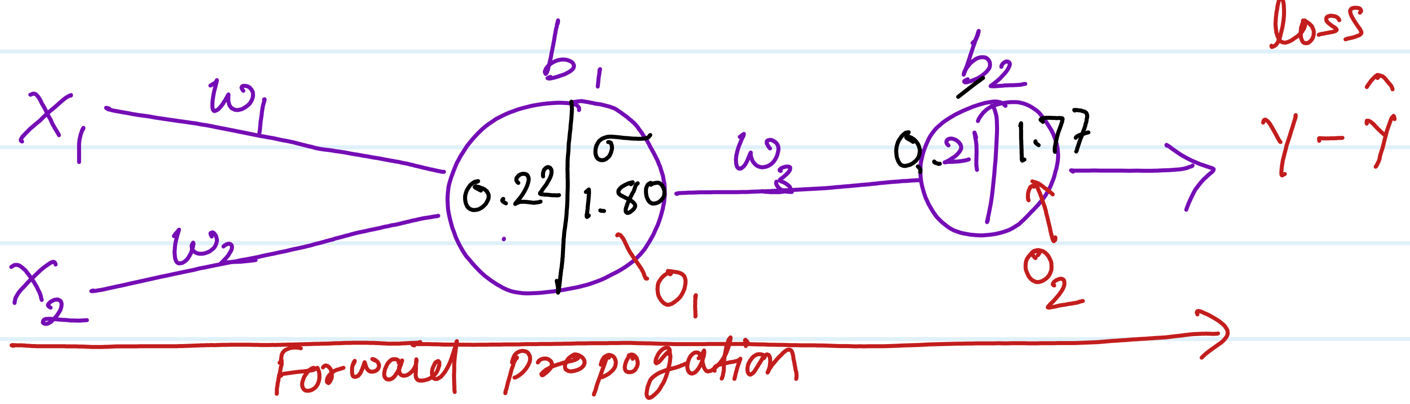
$(y-\hat{y}) = loss$

$z = wx + b$ simple

$z = x_1 w_1 + x_2 w_3 + x_3 w_5 + b_1$

$z = x_1 w_2 + x_2 w_4 + x_3 w_6 + b_2$

4) Forward propogation:-



x_1	x_2	Y
2	5	1

assumed $w_1 = 0.01$, $w_2 = 0.02$, $w_3 = 0.03$, $b_1 = 0.1$, $b_2 = 0.2$

$z = x_1 w_1 + x_2 w_2 + b_1$

$\Rightarrow 2 \times 0.01 + 5 \times 0.02 + 0.1$

$\boxed{\Rightarrow 0.22}$

Activate function

$$\sigma = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-(0.22)}}$$

$$\Rightarrow 0.554$$

$$z_1 = O_1 w_3 + b_2$$

$$= 0.554 \times 0.03 + 0.2$$

$$\Rightarrow 0.21$$

$$\sigma = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-(0.21)}}$$

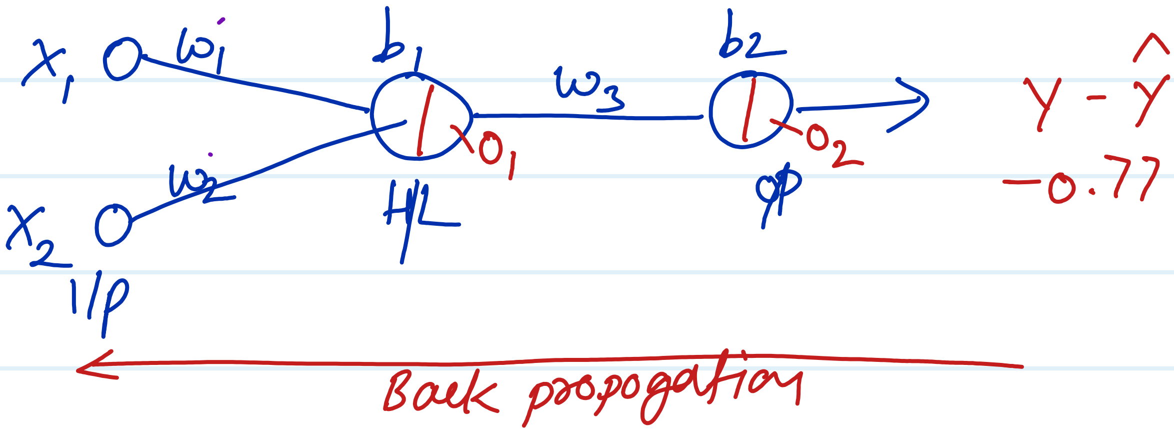
$$\Rightarrow 0.552$$

loss $\quad Y = 1 \quad , \quad \hat{Y} = 0.552$

$$\text{loss} = Y - \hat{Y} = 1 - 0.552$$

$$= 0.45$$

5) Back propogation :-



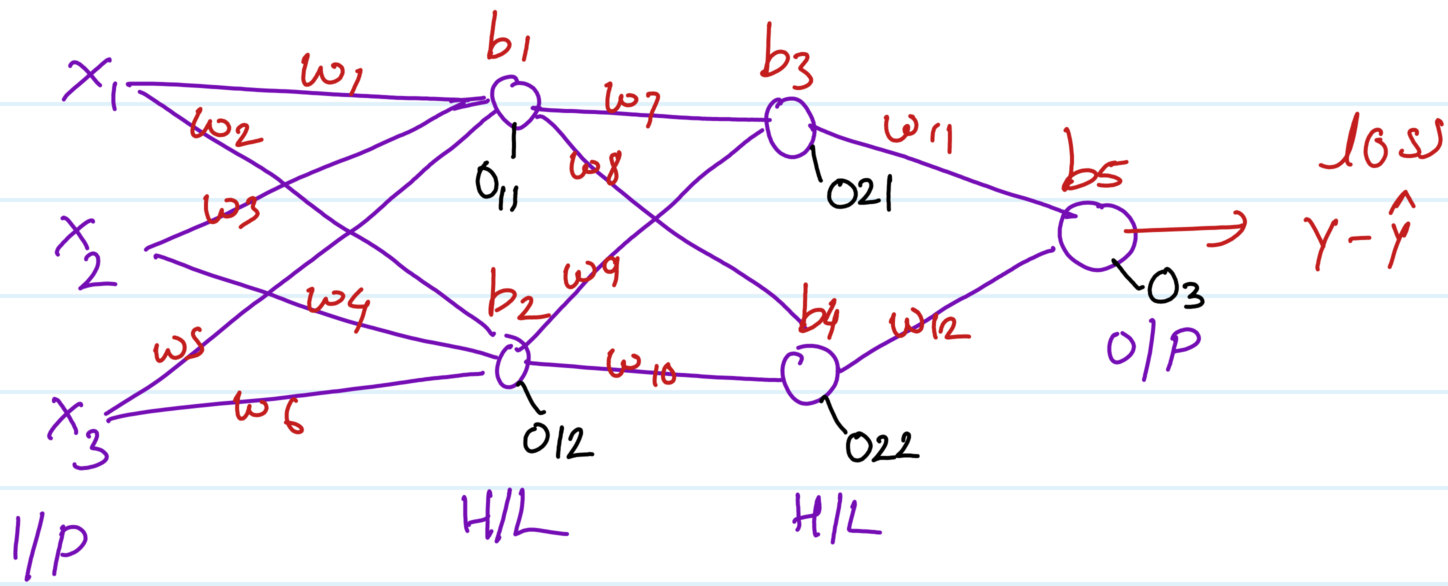
Optimizer (To updat weight and bias)

$$w_{3new} = w_{3old} - \eta \frac{\partial Loss}{\partial w_{3old}}$$

$$b_{2new} = b_{2old} - \eta \frac{\partial L}{\partial b_{2old}}$$

$$w_{1new} = w_{1old} - \eta \frac{\partial L}{\partial w_{1old}}$$

$$\frac{\partial L}{\partial w_{1old}} = \frac{\partial L}{\partial O_2} \times \frac{\partial O_2}{\partial O_1} \times \frac{\partial O_1}{\partial w_{1old}}$$ chain rule



$$w_{11\,new} = w_{11\,old} - \eta \frac{\partial L}{\partial w_{11\,old}}$$

$$w_{9\,new} = w_{9\,old} - \eta \boxed{\frac{\partial L}{\partial w_{9\,old}}}$$

$$\frac{\partial L}{\partial w_{9\,old}} = \frac{\partial L}{\partial O_3} \times \frac{\partial O_3}{\partial O_{21}} \times \frac{\partial O_{21}}{\partial w_{9\,old}}$$

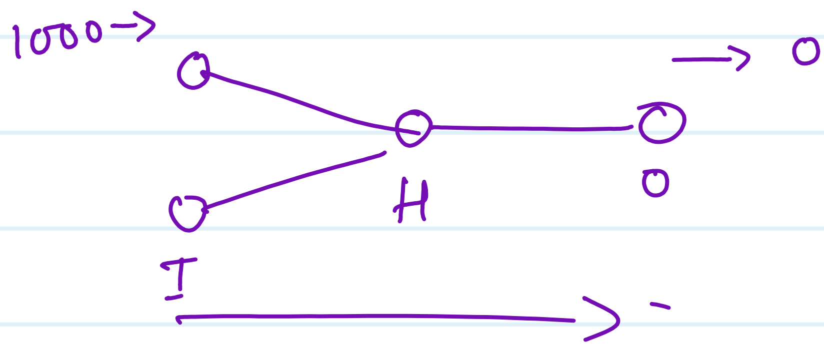
$$w_{3\,new} = w_{3\,old} - \eta \boxed{\frac{\partial L}{\partial w_{3\,old}}}$$

$$\frac{\partial L}{\partial w_{3\,old}} = \left(\frac{\partial L}{\partial O_3} \times \frac{\partial O_3}{\partial O_{21}} \times \frac{\partial O_{21}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial w_{3\,old}}\right) +$$

$$\left(\frac{\partial L}{\partial O_3} \times \frac{\partial O_3}{\partial O_{22}} \times \frac{\partial O_{22}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial w_{3\,old}}\right)$$

7) epoch →

Dataset – 1000



8) Batch size & iteration :-

$$\frac{1000}{\to 40} \Rightarrow 25$$

epoch = 40 iteration

Dataset – 1000

Batch size – 25

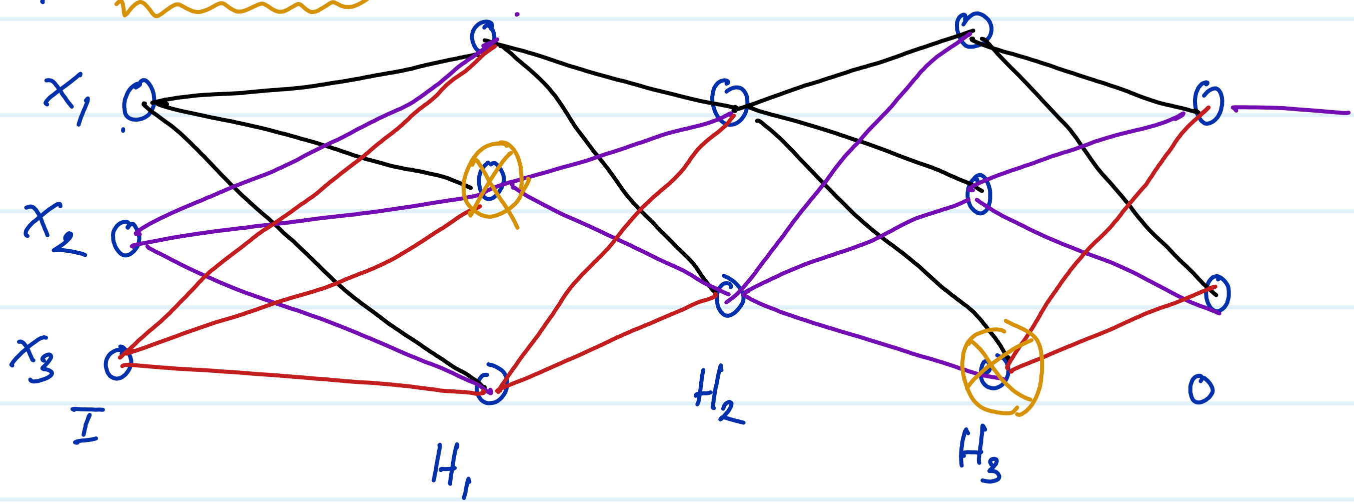
iteration – 40

epoch = 5

total iteration = 5×40

= 200

9) Dropout :- overfitting



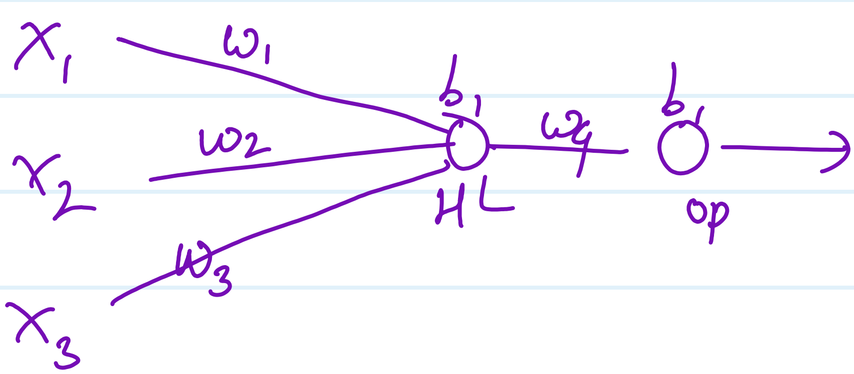
10) Base equation –

$$Z = wx + b$$

$$\boxed{y = mx + c}$$

Support vector machine

weight initializing technique –



① Uniform weight init.

② Xavior & Glorot init. –

③ kaiming He init. –